

MineMind AI — App Development Architecture

Technical Report: Android (.apk) & Windows (.exe) Platforms, Languages & Rationale

CMPDI & Coal India Limited (Ministry of Coal, Govt. of India) | Release Version 2.0 | Sep 2026

1. Android Mobile Application (.apk)

Platform Used: Capacitor 8 (Cross-Platform Native Runtime)

Architecture: Embeds the high-performance React web application inside a hardware-accelerated Android WebView and bridges it directly to the native Android OS through custom Java plugins.

Languages Used & Technical Rationale:

• **TypeScript (.tsx / .ts):**

Role: Screen layouts, touch gestures, offline caching, and UI state. Rationale: 100% code reuse between web and mobile; eliminates rewriting features in Kotlin. Provides strict compile-time type safety preventing runtime crashes on mobile devices.

• **Java (.java) in MainActivity.java:**

Role: Native Android OS bridge (NativeGoogleAuthPlugin). Rationale: Android internal system APIs (such as AccountManager.newChooseAccountIntent() to trigger the system bottom sheet showing only phone Google accounts, and RECORD_AUDIO runtime permissions) are native OS APIs requiring Java.

Key Binary Output: R:\MineMind AI\MineMind-AI-Updated.apk (File Size: 9.5 MB | Status: Ready to Install)

2. Windows Desktop Application (.exe)

Platform Used: Electron 44 + electron-builder (NSIS Installer)

Architecture: Combines a dedicated Chromium rendering engine with a Node.js desktop background process into a single native Windows executable (.exe), running an embedded local offline server on port 3000.

Languages Used & Technical Rationale:

• **TypeScript (.tsx / .ts):**

Role: Desktop workstation window, borehole query bars, charts, and audio waveform meter. Rationale: Strict typing ensures that large geological data sets and heavy PDF reports render cleanly without desktop freezes.

• **JavaScript / CommonJS (.cjs) in electron/main.cjs:**

Role: Desktop lifecycle, process management, and Windows command switches. Rationale: Electron main process executes on Node.js requiring CommonJS to configure native window creation, taskbar icons, file system access, and Chromium media flags.

• **HTML5 Web Audio API (PCM Buffer Engine) in wavAudioRecorder.ts:**

Role: Hardware microphone audio capture. Rationale: Browser cloud speech recognition fails inside Electron because Google restricts it to Chrome. Web Audio API captures raw 16kHz PCM samples, mutes speaker feedback (muteGain = 0), and creates lossless .wav files for Gemini AI transcription.

Key Binary Output: R:\MineMind AI\MineMind-AI-Windows-Updated.exe (Size: 161.9 MB | Status: Ready to Install)

3. Side-by-Side Architectural Comparison Table

Feature / Aspect	Android App (.apk)	Windows Desktop (.exe)	Technical Advantage
Platform Framework	Capacitor 8	Electron 44	100% unified React UI codebase
Packaging Tool	Android Gradle (gradlew.bat)	electron-builder (NSIS)	Standard production distribution formats
Core Languages	TypeScript + Java	TypeScript + CommonJS	Type safety with direct OS capabilities
Final Output File	MineMind-AI-Updated.apk (9.5MB)	MineMind-AI-Windows-Updated.exe (161.9MB)	Direct single-click standalone installers
OS Integration	Android AccountManager API	Windows Node process management	Deep native OS system dialog access
Microphone Pipeline	Capacitor MediaStream	Web Audio API 16kHz PCM + Mute	Lossless audio without feedback screeching
Speech-To-Text AI	Gemini Multimodal Voice	Gemini Multimodal Voice	Direct high-accuracy mining terminology
Execution Mode	Mobile Touch WebView	Embedded Node.js + Chromium Window	Optimized for field phones & office PCs

4. Core Architectural Benefits & Summary

- Single Codebase Strategy: Instead of maintaining separate Kotlin (Android) and C# (Windows) projects, Capacitor and Electron allow MineMind AI to share 95%+ of all React components, styling, and business logic.
- Type Safety via TypeScript: Strict interfaces across borehole logs, DGMS guidelines, and CIL subsidiary records prevent data corruption and unexpected crashes on both phones and workstations.
- Native Java Bridge on Android: Solving the Google login problem by interfacing directly with Android OS AccountManager guarantees that officers choose their genuine device Google accounts.
- Lossless 16kHz PCM WAV Audio Engine: Eliminating reliance on Google Cloud Web Speech in Electron prevents network glitch loops and screeching feedback, delivering fast, grounded speech-to-text with Gemini AI.